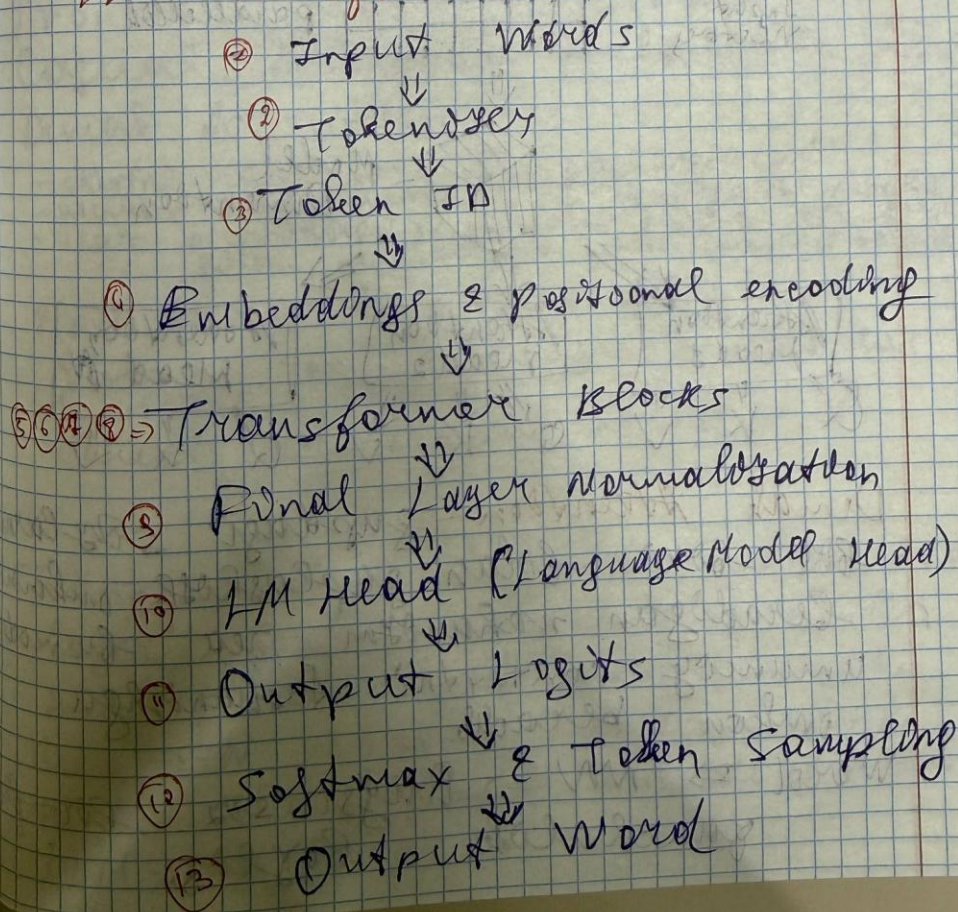


LLM 2-oy 8-Dars Report.

Modellning misal uchun qurilgan
bitta nuqtadagi uchramosdan
tekis shu kishini talminlash uchun
Modular programming va 2-unda
raise value errorlar qilinadi.

LLMning asosiy vazifasi: next
word prediction

LLM Transformer Model Architecture

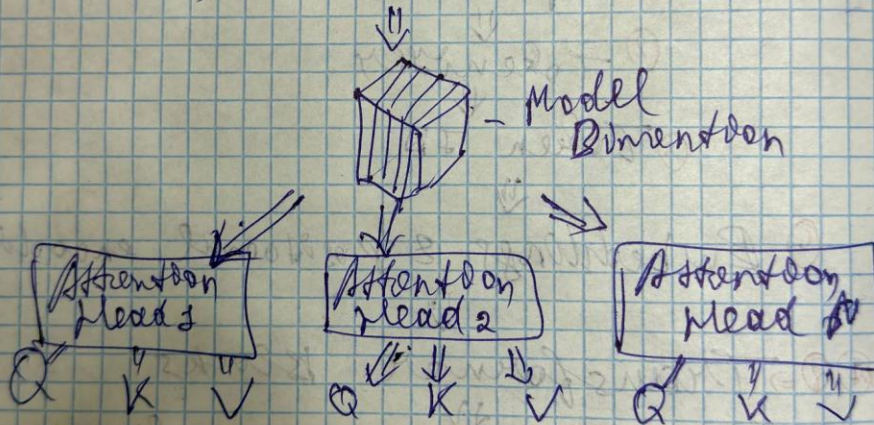


Residual - kelayotgan bloklarni boglab beradi
 Attention

Word $\xrightarrow{\text{BPE}}$ Token $\xrightarrow{\text{Vocabulary}}$ TokenID $\Rightarrow$ Embeddings $\xrightarrow{\text{FFN}}$ Trans Attention
 shu joyda head!

Casual Multi-head Attention

Input vectors parallellok



Demak Attention - gapdagi soxlarni bir-biri bilan paplashish imkonini beradigan mexanizm va bu mexanizm umumiy kontekstni hisoblashga imkon beradi

Word $\Rightarrow$ NN $\rightarrow$ Ragsi soz bir-biri ganchalok boglag

Casual Self Attention - Self Attention
juni ra unding asosiy funksiyasi

Simple (global) Attention - reading
files, images, code.

generating text, chatting/math,
reasoning uchun - Casual Attention

allows step by step sequential
thinking and token generation
without cheating.

Frontier modellar simple ra
casual attention suriladigan fayd
Amaliy Attention

- so'zni bir biri bilan gaplashadi
- ma'ni ma'nosizlik bilan ma'ni uchun hisob
- near erashadi, zamonaviy tashabbi qo'riq qo'riq
bilan nima uchun qiziqish uchun beradi.

2 xil turdagi bir tashabbi

Global 360° kamerga o'xshab har
bir soni ko'radi

Casual esa o'zini ko'radi.

FFN

Transformerlar korgandan keyin 2 to
layer bir 1 to Attention bilan

2 to FFN - feed forward network.

Be FFN modelni memory banki
jaini eslab qolish uchun ishlatiladi
LLM ning noga

FFN vs Attention parallel sequential vs parallel

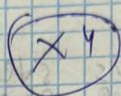
The FFN Layer

Input Tensors



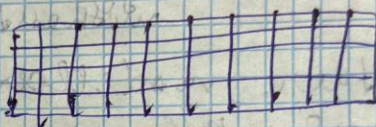
content size 384
(256 x 384)

Step 1: up scale



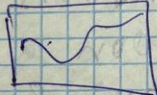
4x parallel

A hidden = 384

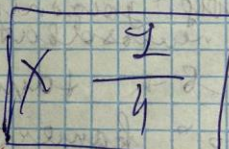


content size 1536

Gelu



Step 2: Re-compress



n-embed = 384



Compression back to model dimension

Attention - by Dynamic (Transformer)

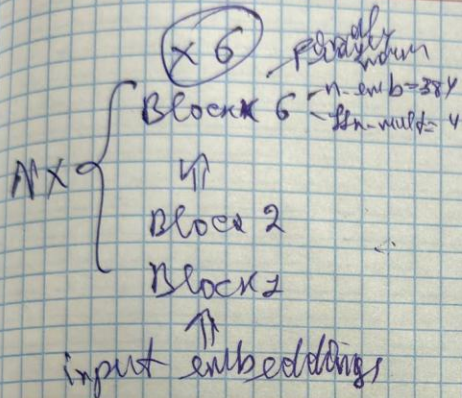
FFN - by static (Transformer)

Attention - by learn for - long and short range

el yoko

FFN (MLP) - Faktlar, ma'lumotlar
va birlamchi saqlash hamda qayta
ishlash.

Transformer Block



Transformer Blocks (n-layer=6)



Feed-Forward



Residual (+)



Attention

Transformer asosiy bino.

Residuals - ma'lumotni yig'ib olishda
qo'shimcha qo'shiladi.

Model - yig'ib olish jarayoni, barchasini
bir joyga yig'ib olish.

Test - LLM Development: Pre-training
sanity checks (review slide)

1. Initial Loss check (uniform bias check)

2. Causality verification (the anti-cheat
check)

Test - training dan oldin matematik
nazorat qilish

Initial loss - modelning dastlabki
natijalari va adabatlari

Causality - Verification - casual self attention
qo'shiladi.